10.11.2025

EEE412 Microwave Electronics- Homework 3

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"I completed this homework assignment independently."

Part 1 and 2

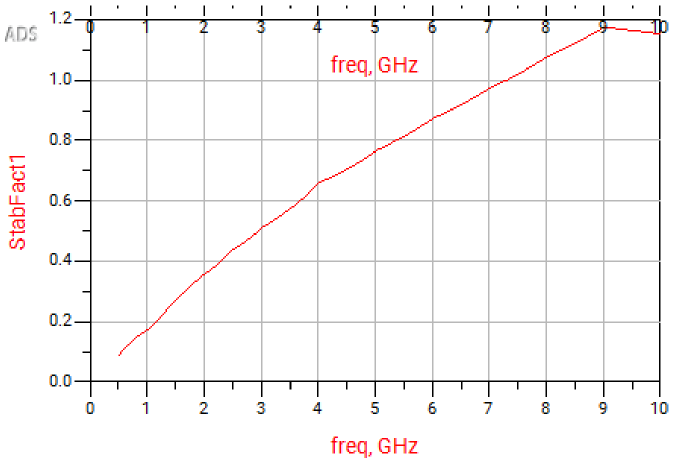

Figure 1: Stability Factor (K)

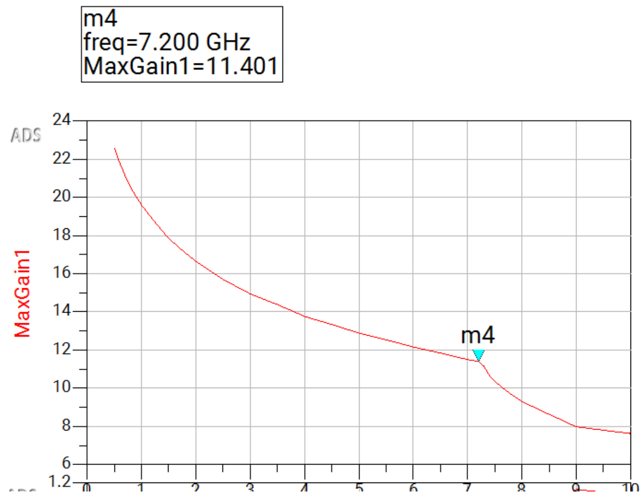

Figure 2: Maximum Available Gain (MAG)/Maximum Stable Gain (MSG) Plot

Note that kink value occurs at 7.2MHz, this, point is where MSG end and MAG starts also, this corresponds to the point where stability factor reaches above 1 (K>1).

Part 3

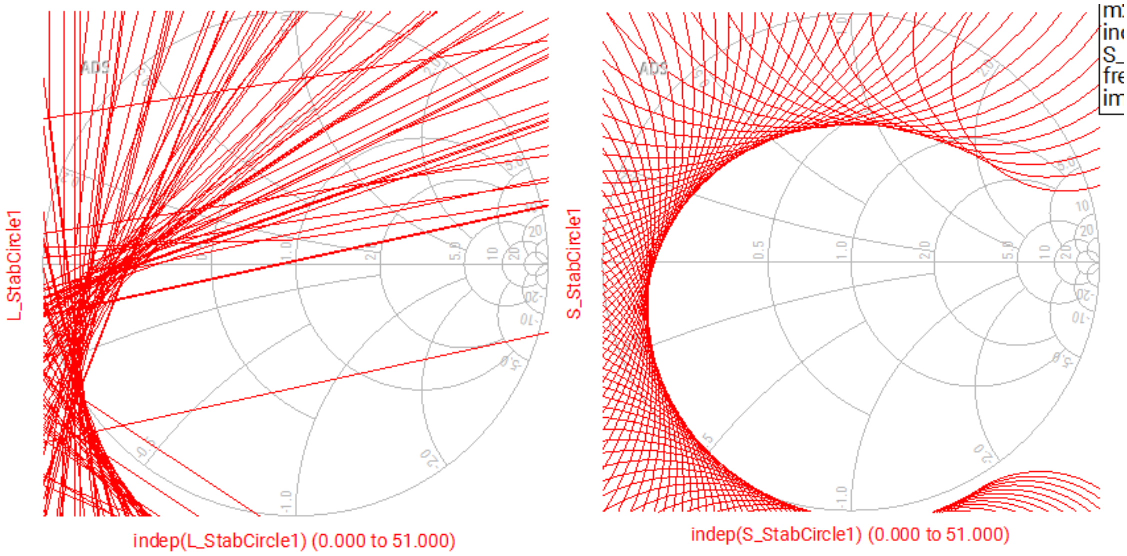

Figure 3 and 4 : Load Stability Circles and Source Stability Circles Respectively

Part 4

We choose the input impedence value as Z=-24.49-j35.12, see Figure 5 for the circuit arrangement and Figure 6 to see S22 goes outside of the unity circle.

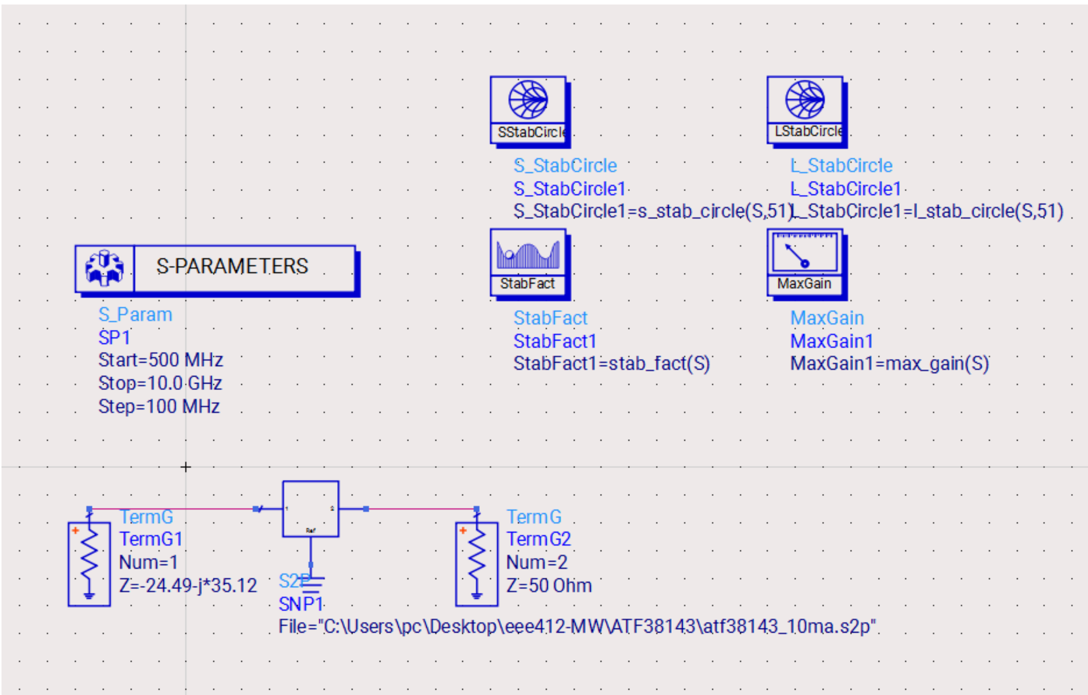

Figure 5: Terminated Input to Push the s22 to Unstable Region

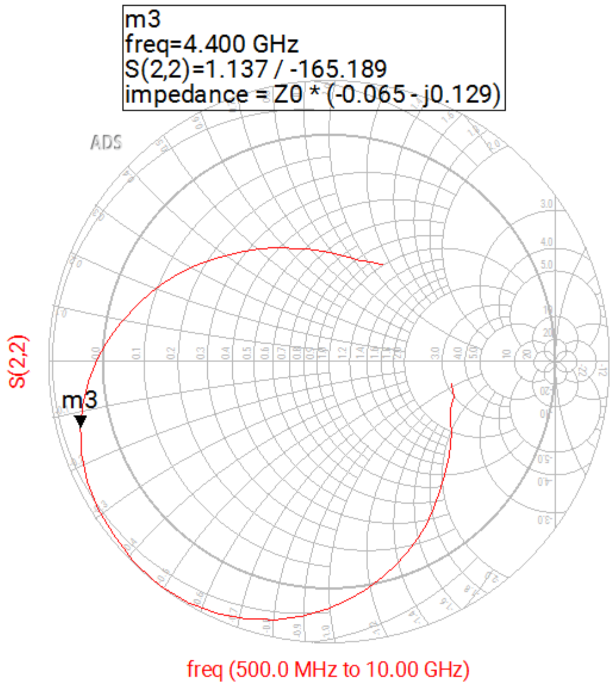

Figure 6: S22 Result of Figure 5

Part 5

I picked the frequency f1= 7.2MHz, I add a resistive network in order to stabilize the transistor at all frequencies. After tuning, below values for my netwrok occured. See below Figures to check MAG and Stability Factor K for the given frequencies. MAG values are really close to the previous plot, and Min of K stays above 1.

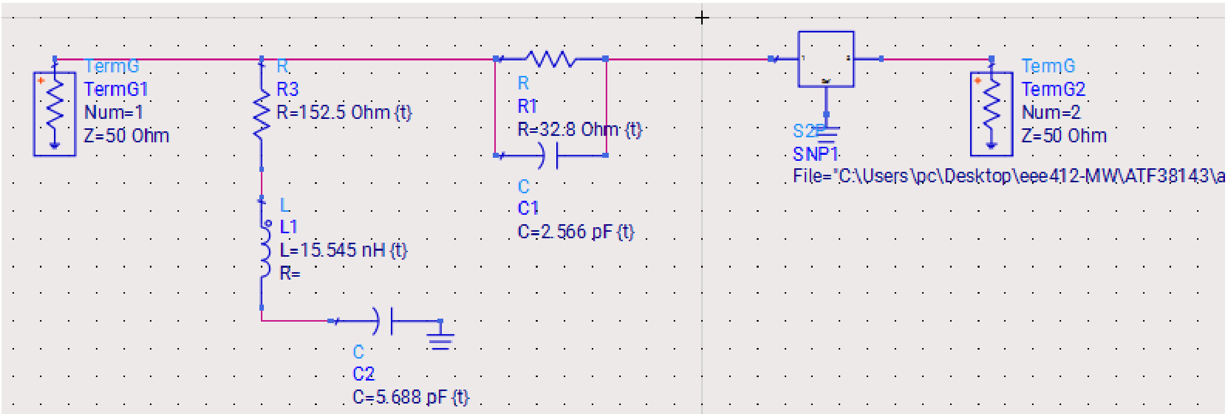

Figure 7: Resistive Network Addition to my Input

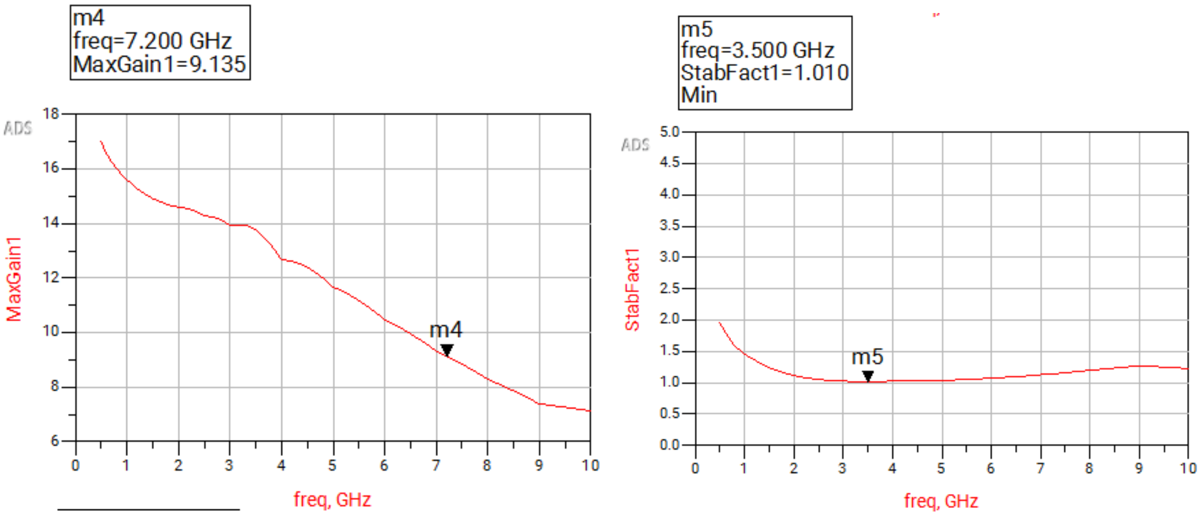

Figure 8 and 9: Maximum Gain Plot and Stability Factor (K) Plot

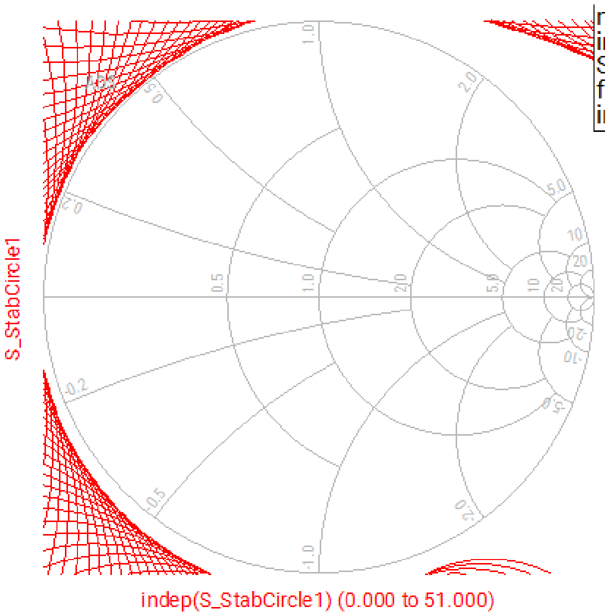

Figure 10: Stability Circles Stays out of the Smith Chart

Part 6

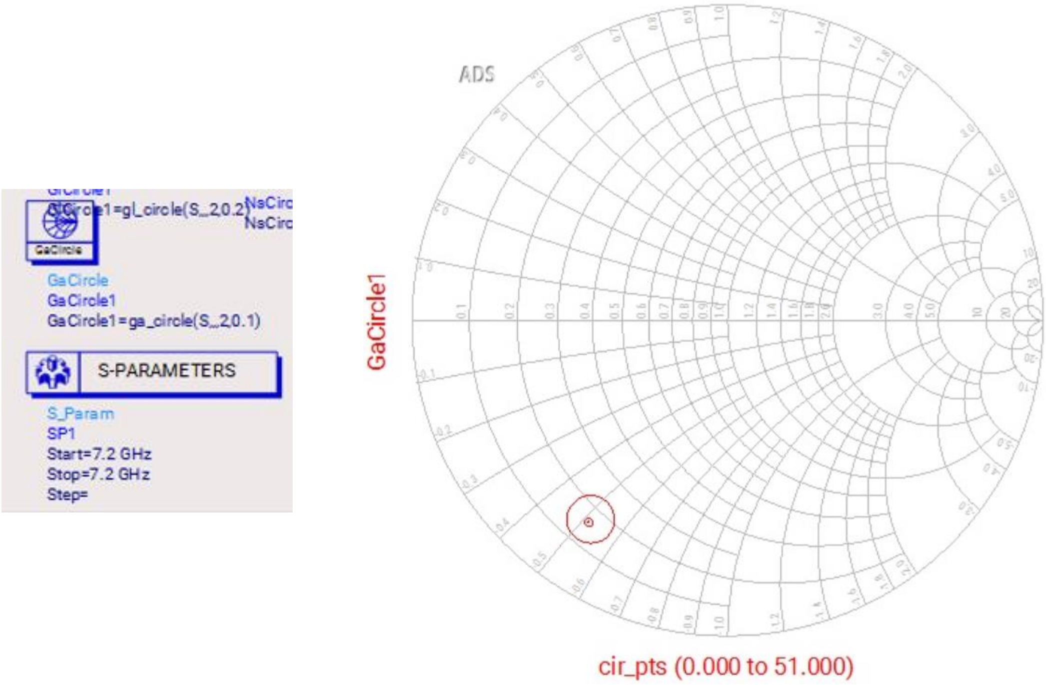

Figure 11: (MAG) and a circle at MAG-0.1 in dB

Part 7

For the input matching we should go with a shunt capacitor then a series inductor, can be seen again the Ga Circle position with respect to the center of the plot. The shunt capacitor and series inductor was added to the input port, then tuned (See part 13 to see my centeralized Ga Circle).

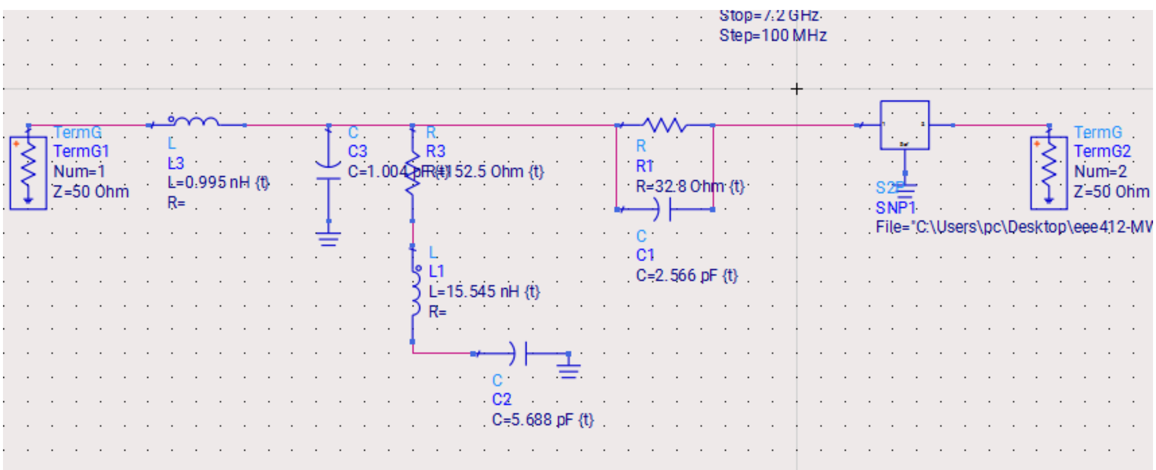

Figure 12: Tuned Circuit that Centeralizes Ga Circle

Part 8

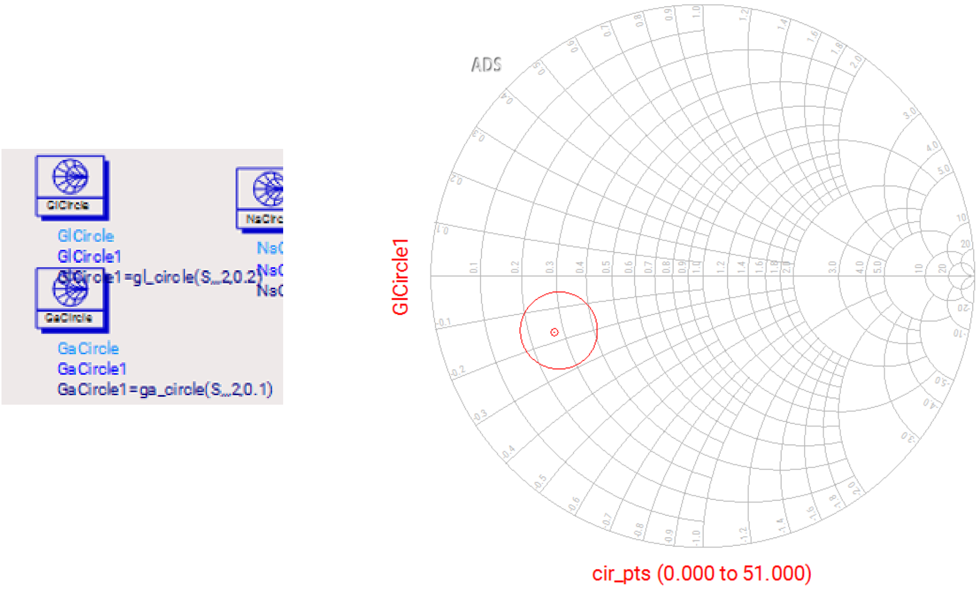

Figure 12: Best Load impedance Values with a 0.2dB Gain Reduction.

Part 9

The following output matchint network is added so that Gl circle will be centralized, we will need first a series L then a shunt C. Then we tune it.

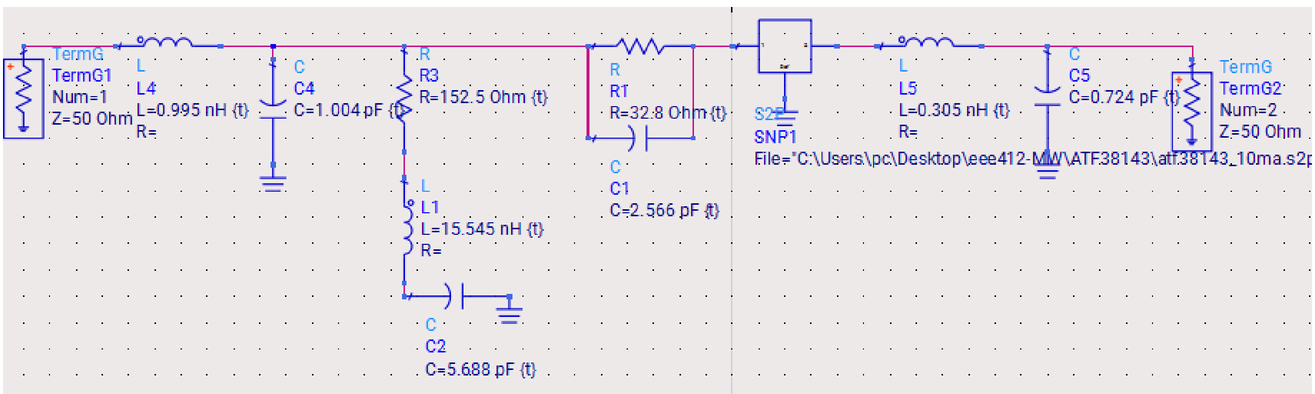

Figure 13: Tuned Circuit that Centeralizes Gl Circle

See from the below Figures now, S21 equals to Max Gain = 9.135dB at that frequency (7.2MHz).

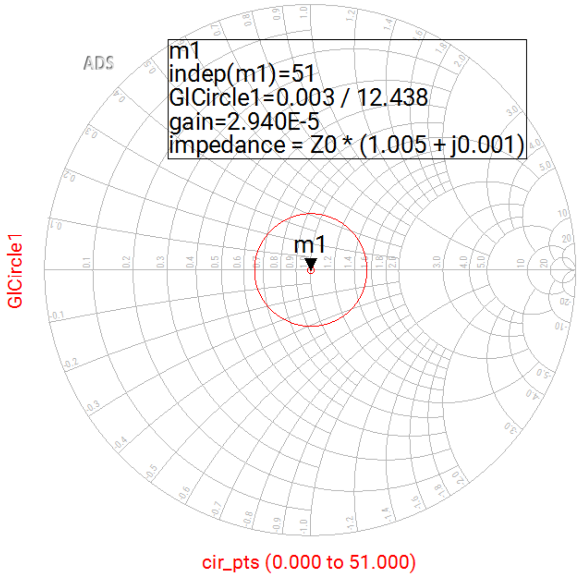

Figure 14: GI Circles Centralized

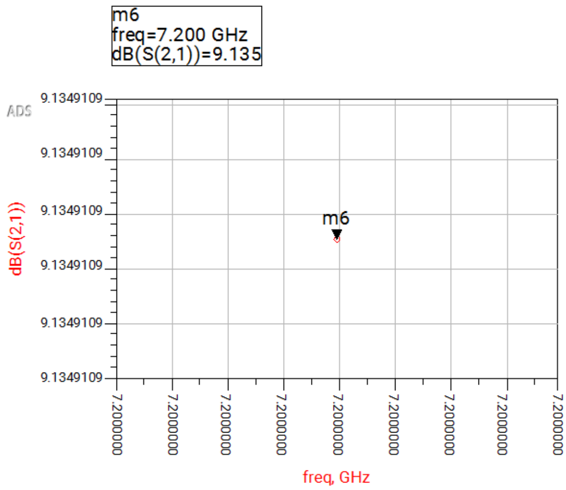

Figure 15: S21 equals to 9.135dB

Part 10

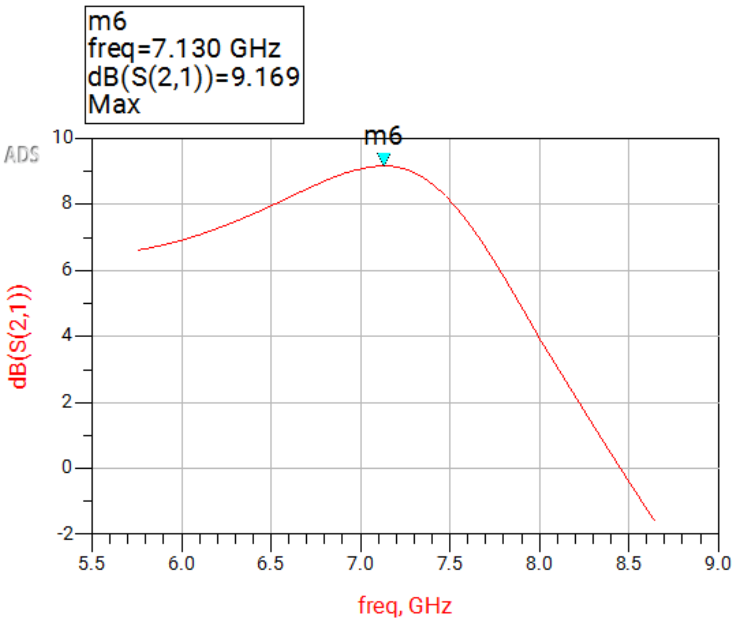

Figure 16: as a Function of Frequency

Part 11

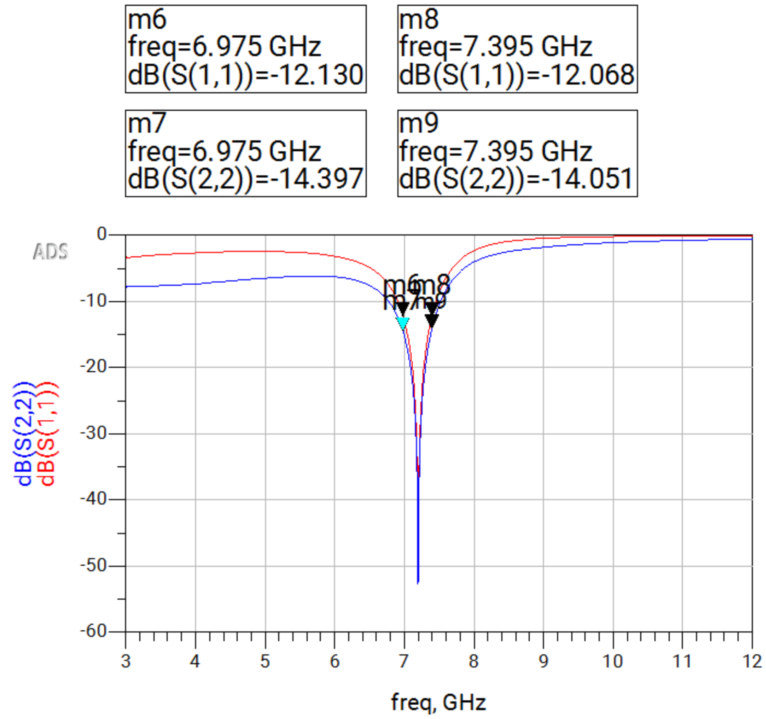

Figure 17: S11 and S22 Values Both Lower Than -12dB

Part 12

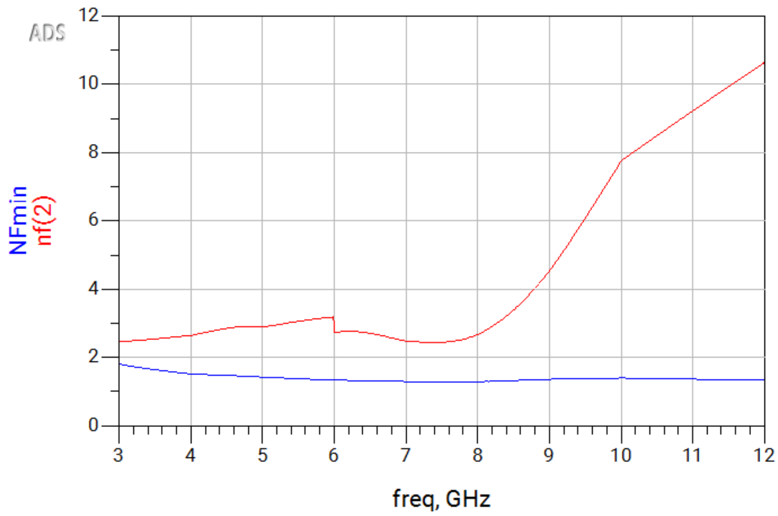

Figure 18: NFmin Minimum Noise Figure and nf(2) Comparison

Part 13

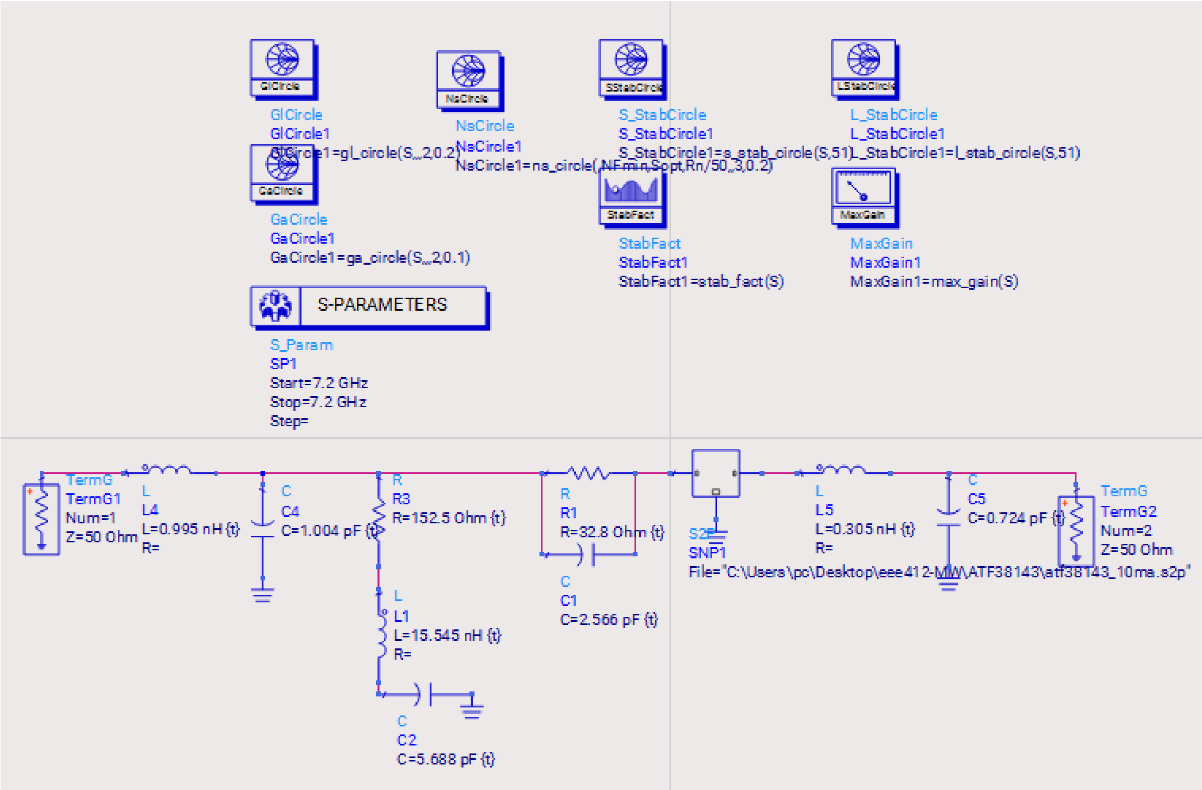

Figure 19: Circuit and Setup for Part 13

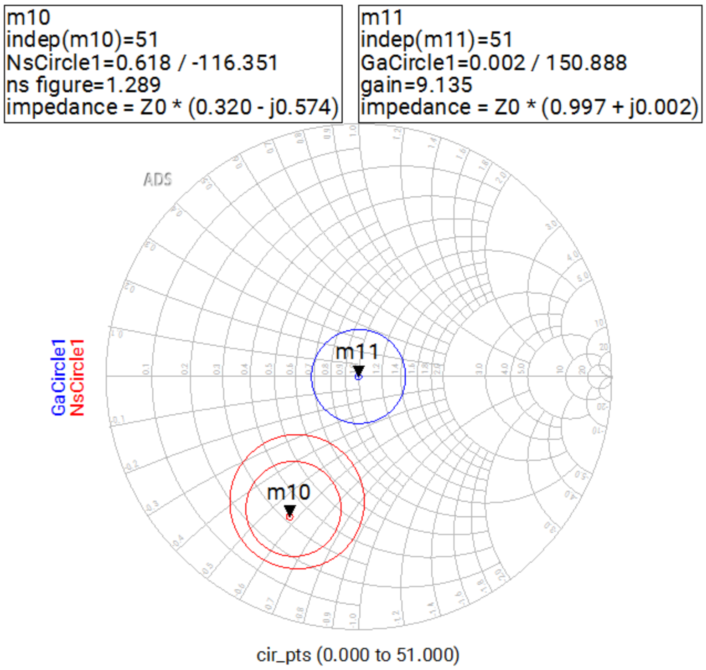

Figure 20: NFmin, NFmin+0.2, NFmin+0.4 dB at f1 with a Ga Circle

Part 14

The below circuit is both Centralizes the Noise Circle for f1 to the center and also the 0.2dB noise figure circle touches the center while Ga is reduced as little as possible. When the input matching network is varied, the Ga circles will no longer be centered. Therefore, a compromise exists between maximum gain and minimum noise figure.

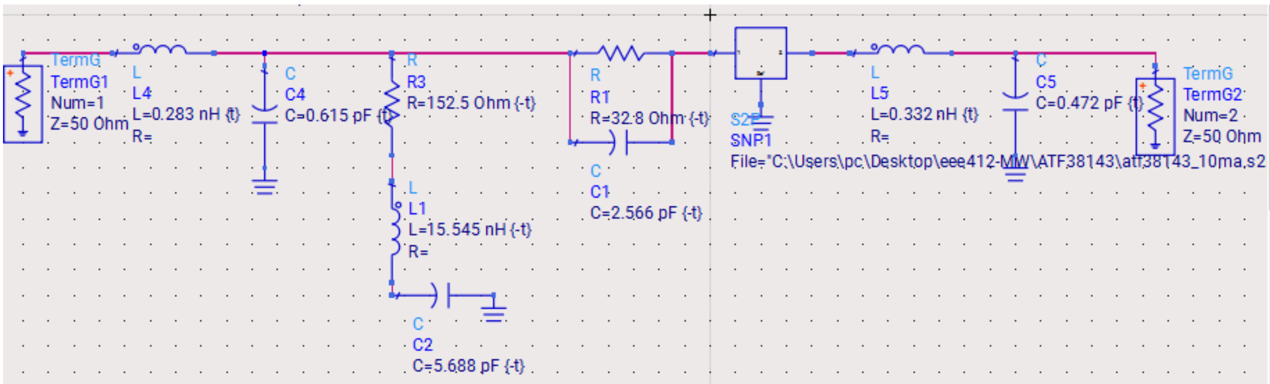

Figure 21: Tuned Circuit for Noise Circle 0.2dB Touches to center

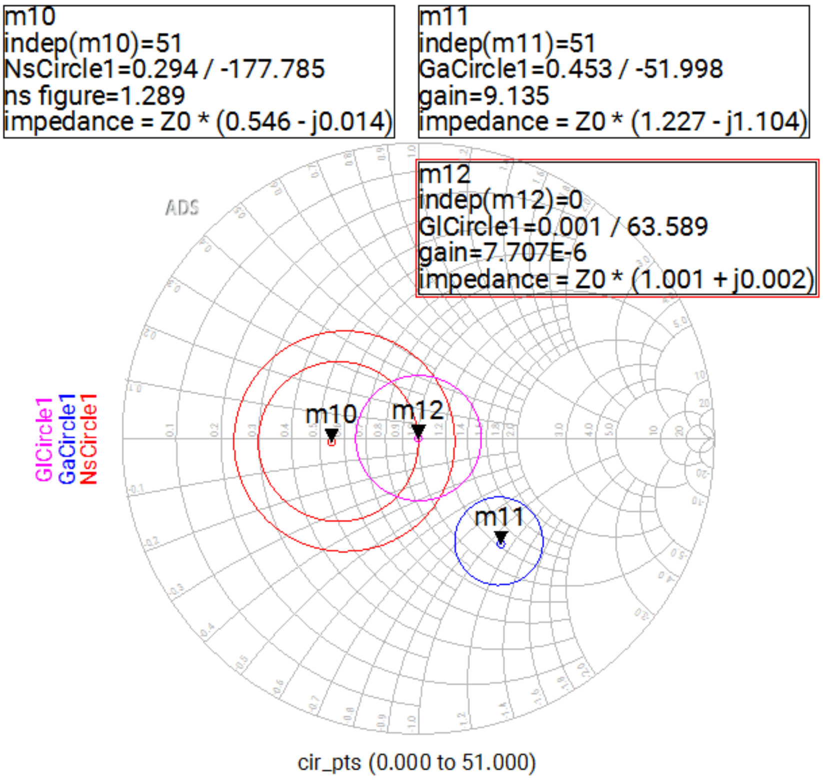

Figure 22: Plots of Figure 21 and Part 14

Part 15

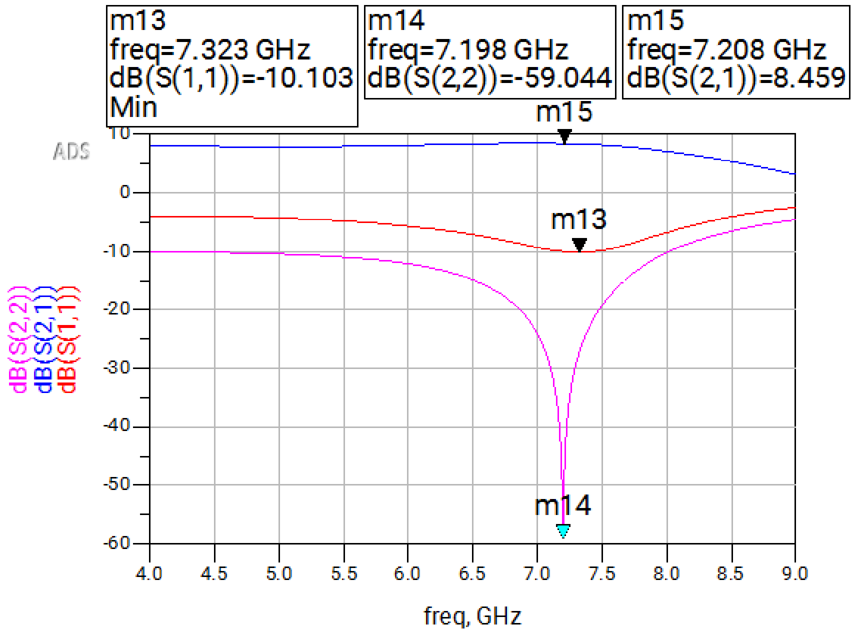

Figure 23: Plots of S11, S22, S21

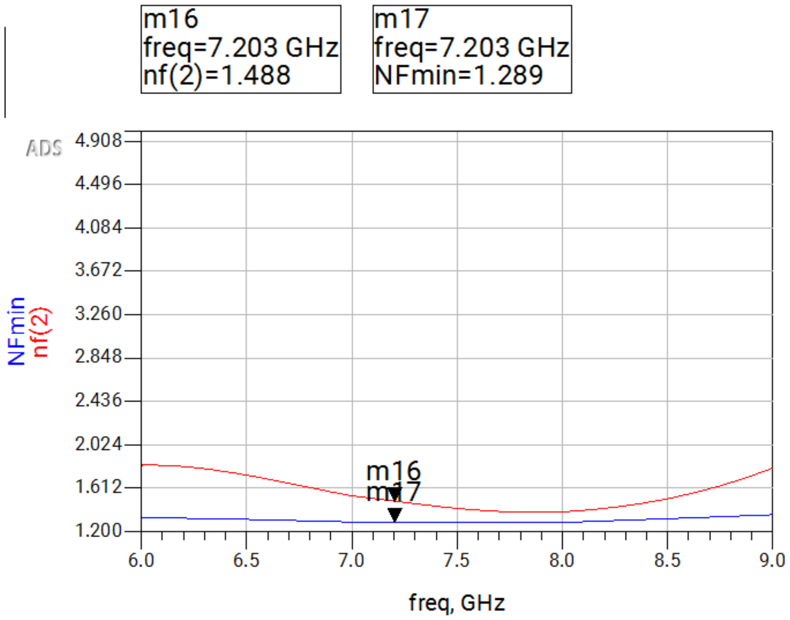

Figure 24: Plots of Noise Figure (F) and Fmin

Part 16

Using this inductor, it is possible to bring the noise figure and the Ga circles closer together. If the inductor is made larger, MAG will reduce. Since the two-port is modified, check that the stability factor is still above unity at all frequencies. Insertion of the small inductor may also help in stability.

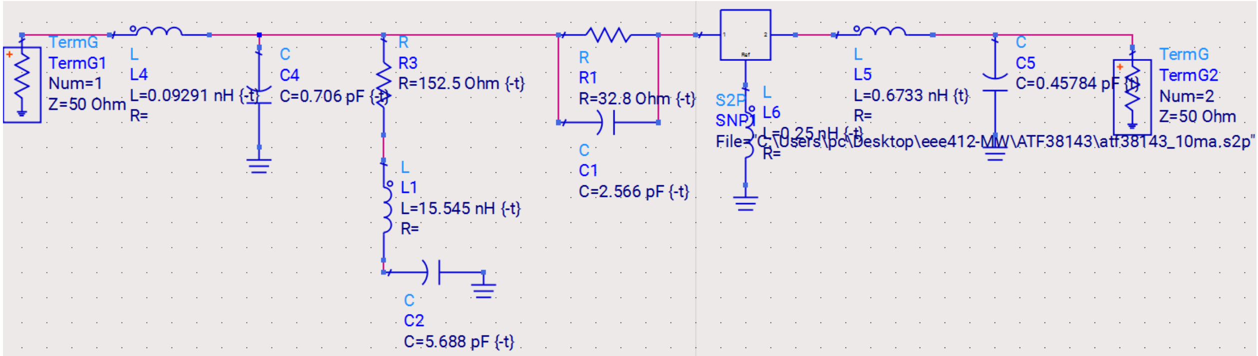

Figure 25: Overall Tuned Circuit After Inductor to the Source of the Transistor

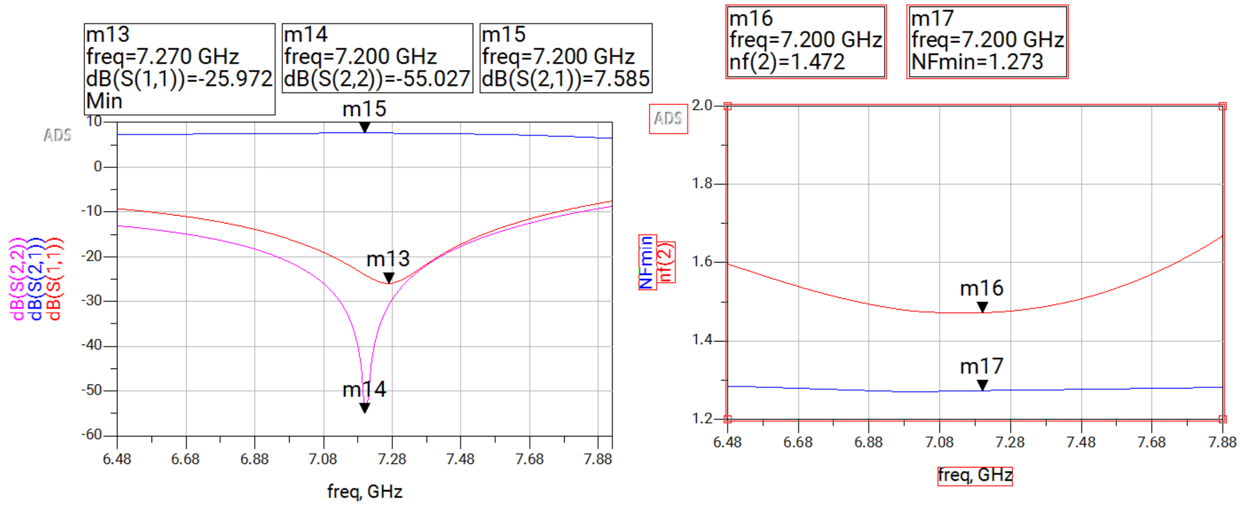

Figure 26: New Optimized Results